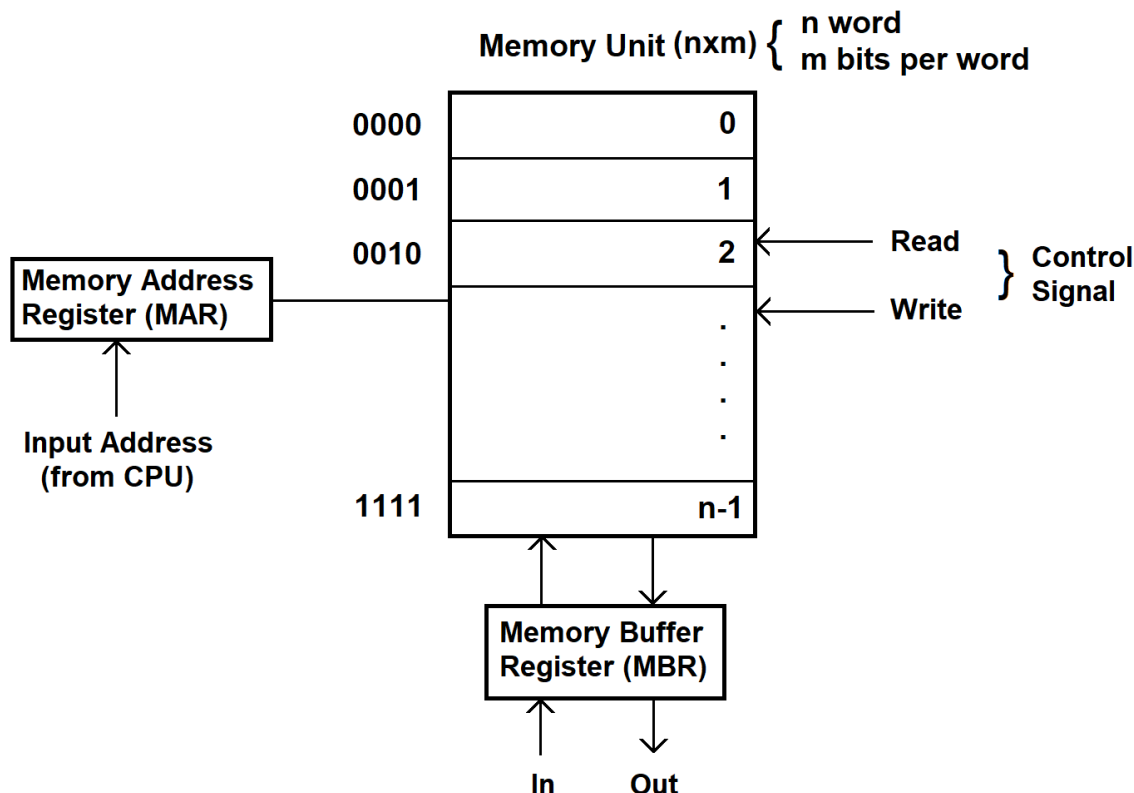


Lecture 10 - Memory

Registers: A register is a group of flip-flops combined together to store multiple bits. An n-bit register contains n flip-flops and can store n-bit information.



Memory Unit: A memory unit stores binary information in groups of bits called words.

Memory Address Register (MAR): To communicate with a memory word, its address is transferred to the memory address register. The internal circuit then opens the path to select that word. A n-bit address register can specify up to 2^n addresses.

Memory Buffer Register(MBR): Before writing and after reading information are kept in Memory Buffer registers. In each case, the address register specifies location of writing or reading.

Control Signal: The control lines Read and Write (usually combined into a single control line Read/Write) specifies the direction of transfer of the data.

Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

Why do we need the MBR? Why can't the CPU directly send the data to the memory unit?

The MBR acts as a buffer between the CPU and memory, ensuring data is temporarily stored and synchronized during read/write operations. Without it, direct communication would be unreliable and harder to control.

Example: A memory unit of 1024 words with 8 bit/word needs, 10 address line (i.e. address register must contain 10 FF) and buffer register must have 8 FF to store words transferred in and out of it.

Memory Read Operation:

CPU issues a read request to the memory unit to fetch data from a specific memory location

Steps:

1. CPU sends the memory address from where data will be read into the MAR.
2. The memory unit activates the corresponding memory location.
3. The read control signal will be activated. (Read = 1 or RW = 0)
4. Data at the address is fetched and placed in the MBR.

Memory Write Operation:

CPU issues a write request to the memory to write data to a specific memory location

Steps:

1. CPU sends the memory address where data will be written into the MAR
2. Places the data to be written in the MBR
3. The write control signal will be activated. (Write = 1 or RW = 1)
4. Memory locates the given address and stores the data in it.

Formulas to remember:

What is the capacity of the memory?,

$$\text{Capacity} = \text{no. of word} \times (\text{bit/word})$$

How many words are there?

$$\text{No. of word} = 2^{\text{(no. of address lines)}}$$

Sometimes, Number of Address lines is referred to only Address Lines

How many bits are required in MAR?

$$\text{Address lines} = \log_2(\text{No of word})$$

Unit Conversion:

8 bit = 1 byte

1024 byte = 1 KB

1024 KB = 1 MB

1024 MB = 1 GB

1024 GB = 1 TB

1024 TB = 1 PB

In short form, Bits are represented in small letter b. Example: 1 Kb = 1 Kilobits

And Bytes are represented in capital letter B. Example: 1 MB = 1 Megabyte

Types of Semiconductor Memories

ROM – Read Only Memory - a type of memory that cannot be written, can only be read. Contents determined at manufacture time

RAM – Random Access Memory - Memory that can be both read and written during normal operation. Also, contents are volatile (will be lost on power off).

Two types of RAM:

Static RAM (SRAM):

Uses flip-flops as the memory cells, Complex Circuitry.

No need to refresh constantly

Typically the type of RAM used primarily as cache memory.

Dynamic RAM (DRAM):

Use capacitor charges to represent data.

Though simpler in circuitry, they have to be constantly refreshed.

Most main memory is dynamic RAM

Dynamic RAM (DRAM) Types:

SDRAM – Synchronous dynamic RAM – operates synchronously with system clock and data bus. Can handle 100MHz to >800MHz.

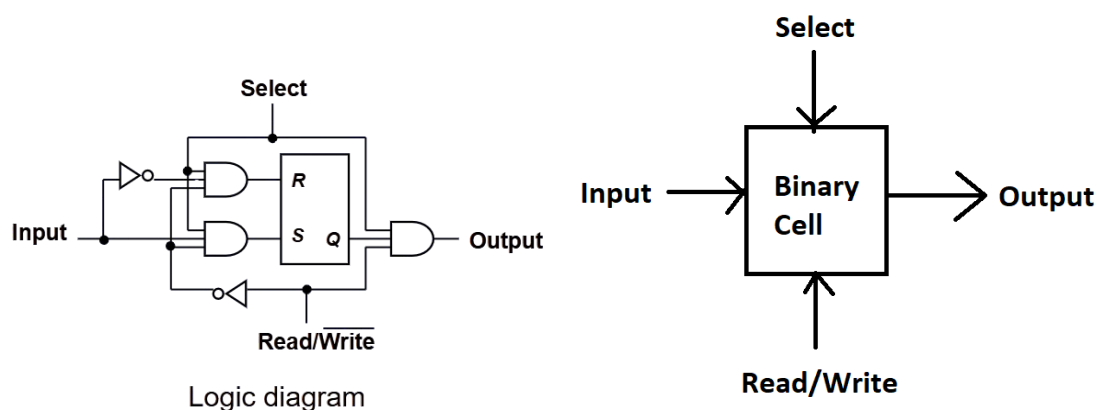
DDR – Double Data Rate – can transmit data on both edges of the clock. DDR3 = Double Data Rate 3rd generation.

QDR – Quad Data Rate – twice as fast as DDR

Typically we use DDR3, DDR4 & DDR5 RAM as main memory in our windows PC and Static RAM for L1, L2 and L3 cache memory.

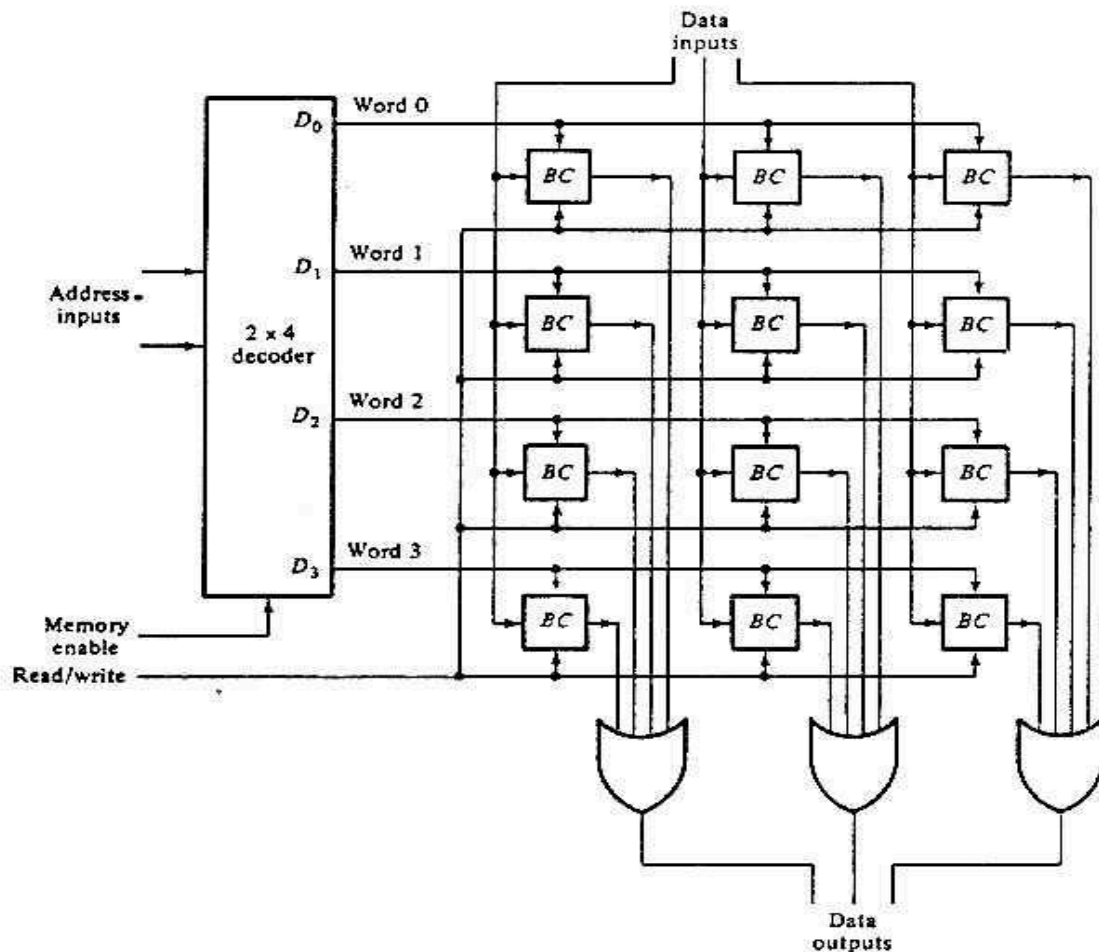
Binary Cell: An unit of computer storage that can store one bit of information.

A single memory cell of the static RAM has the following logic and block diagrams.



Logic construction of a 4 x 3 RAM (with decoder and OR gates):

Here 4 means 4 words. 3 means 3 bit per word



Practice Problems:

- How many address lines do we need for a 64 MB RAM with 16 bit/words?
- Design a 2048x32 RAM showing internal details.
- Design a 4096x16 RAM showing internal details.
- What is the capacity for a $2^{16} \times 16$ RAM in Gigabytes?
- Draw the block diagram of a binary cell.
- Draw and explain the functionalities of MAR and MBR.
- What type of information can you get from the “2048x32” part of a 2048x32 RAM?
- How many types of RAM are there?
- What is static RAM?
- What is dynamic RAM?
- Convert 8,796,093,022,208 Bits to Gigabytes.
- Convert 3 Terabytes to bits.